PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Clinical outcome measures and their evidence base in degenerative cervical myelopathy: a systematic review to inform a core measurement set (AO Spine RECODE-DCM)
AUTHORS	Yanez Touzet, Alvaro; Bhatti, Aniqah; Dohle, Esmee; Bhatti, Faheem; Lee, Keng Siang; Furlan, Julio C; Fehlings, Michael G.; Harrop, James; Zipser, Carl Moritz; Rodrigues-Pinto, Ricardo; Milligan, James; Sarewitz, Ellen; Curt, Armin; Rahimi-Movaghar, Vafa; Aarabi, Bizhan; Boerger, Timothy; Tetreault, Lindsay; Chen, Robert; Guest, James; Kalsi-Ryan, Sukhvinder; McNair, Angus; Kotter, Mark; Davies, Benjamin; RECODE-DCM Steering Committee, AO Spine

VERSION 1 – REVIEW

REVIEWER	Atsushi KimuraKimura Jichi Medical University, Department of Orthopaedics
REVIEW RETURNED	14-Oct-2021

GENERAL COMMENTS	This study systematically reviews the measurement properties of outcome measures currently used in the assessment of degenerative cervical myelopathy (DCM) for clinical research. This information will be useful for clinicians to select adequate outcome measures for the treatment of DCM. This paper is well-organized and seems to be sound in the science. I have only a minor comment. 1. In supplemental table 2, the correspondence between a study name and outcome measures seems to be obscure. Drawing additional lines may improve the comprehensibility of the table.

REVIEWER	Ximin Li
REVIEW RETURNED	27-Oct-2021

GENERAL COMMENTS	This manuscript aims to evaluate the measurement instruments used in assess DCM based on Consensus-based Standards for the selection of health Measurement Instruments, and made recommendation on instrument to improve consistency in measurement and reporting in DCM research. The author did a comprehensive work in searching and screening the eligible studies, evaluate the quality through different aspects. There are some aspects I hope to be clarified: For each study, how many reviewers assessed the quality, feasibility and interpretability? And if the assessment is inconsistent between raters, what procedure were taken? How the quality of evidence is

	defined In Quality assessment section, could the author add a sentence or two to briefly explain mGRADE approach or add citation to it. In Data analysis section, could the author briefly elaborate the definition or the criteria for “sufficient”, “insufficient”, “high quality”, “low quality”. Since the interpretability plays an important role in recommendation. Could the author elaborate the criteria or standard for interpretability assessment? The legend of the color coding is missing for the heatmap (Fit 2) In table 2, the definition of interpretability is the same as responsiveness.

REVIEWER	Murali Sayana Our Lady of Lourdes Hospital
REVIEW RETURNED	31-Oct-2021

GENERAL COMMENTS	The fact there is no Type A recommendation suggests that COSMIN criteria may not be fit for purpose for DCM??

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. Atsushi KimuraKimura, Jichi Medical University

Comments to the Author:

This study systematically reviews the measurement properties of outcome

measures currently used in the assessment of degenerative cervical myelopathy

(DCM) for clinical research. This information will be useful for clinicians to select adequate outcome measures for the treatment of DCM. This paper is well-organized and seems to be sound in the science. I have only a minor comment.

Our response: Thank you very much for your appreciation.

1. In supplemental table 2, the correspondence between a study name and outcome measures seems to be obscure. Drawing additional lines may improve the comprehensibility of the table.

Our response: Thank you for this suggestion. To improve the comprehensibility of Supplementary table 2, we have drawn the additional lines between rows, as well as separated the items in the last two columns by new lines instead of commas. This has been re-uploaded as a new file (‘Supplementary material-revision).

Reviewer: 2

Dr. Ximin Li

Comments to the Author:

This manuscript aims to evaluate the measurement instruments used in assess DCM based on Consensus-based Standards for the selection of health Measurement Instruments, and made recommendation on instrument to improve consistency in measurement and reporting in DCM research. The author did a comprehensive work in searching and screening the eligible studies, evaluate the quality through different aspects.

Our response: Thank you very much for your appreciation.

There are some aspects I hope to be clarified:

For each study, how many reviewers assessed the quality, feasibility and interpretability? And if the assessment is inconsistent between raters, what procedure were taken? How the quality of evidence is defined

Our response: Thank you for these comments. We have added a sentence to the “Quality assessment” and “Data extraction” sections that specifies how many reviewers assessed the quality, feasibility, and interpretability of the studies, and how disagreements were resolved.

In Quality assessment section, could the author add a sentence or two to briefly explain mGRADE approach or add citation to it.

Our response: Thank you for this comment. We have added the citation to the COSMIN guidelines that provides a detailed explanation of the mGRADE approach taken.

In Data analysis section, could the author briefly elaborate the definition or the criteria for “sufficient”, “insufficient”, “high quality”, “low quality”.

Our response: Thank you for this comment. As with the “Quality assessment” section, we have added to the “Data analysis” section the citation to the COSMIN guidelines that provides a detailed elaboration on the definitions and criteria of “sufficient”, “insufficient”, “high quality”, “low quality”.

Since the interpretability plays an important role in recommendation. Could the author elaborate the criteria or standard for interpretability assessment?

Our response: Thank you for this suggestion. We have added a sentence to the “Quality assessment” section that elaborates the a priori criteria we have used for the interpretability, and feasibility, assessments. We have also added a parenthesis to the “Results” section that reiterates the criteria for interpretability, to reemphasise this important point before elaborating on it in the “Discussion”. Furthermore, citations to the COSMIN guidelines, which expand on these criteria, have been added to these sections, as well as to Tables 4 and 5, where these findings are presented.

The legend of the color coding is missing for the heatmap (Fit 2)

Our response: Thank you for this comment. We have added the color coding legend to Figure 2 and re-uploaded it as a new file (‘Figure 2-revised’).

In table 2, the definition of interpretability is the same as responsiveness.

Our response: Thank you for this comment. We have added the correct COSMIN definition of interpretability to Table 2.

Reviewer: 3

Dr. Murali Sayana, Our Lady of Lourdes Hospital

Comments to the Author:

The fact there is no Type A recommendation suggests that COSMIN criteria may not be fit for purpose for DCM??

Our response: Thank you for this comment. On the one hand, the COSMIN criteria are robustly developed and are very relevant for the advancement of high-quality measurement instruments. On the other, we do not surmise that, at this point in time, category A instruments are *necessary* for the practical advancement of the DCM field, mainly because absence of high-quality evidence does not equal presence of poor-quality evidence. This question posed by the reviewer is indeed an important one, and we have accordingly clarified the wording of our answer and reasoning in the last paragraph of the “Results” section, as well as in the third and fourth paragraphs of the “Discussion” section, where we elaborate on this in greater detail.

VERSION 2 – REVIEW

REVIEWER	Ximin Li
REVIEW RETURNED	21-Dec-2021

GENERAL COMMENTS	This manuscript aims to evaluate the measurement instruments used in assess DCM based on Consensus-based Standards for the selection of health Measurement Instruments, and made

	recommendation on instrument to improve consistency in measurement and reporting in DCM research. The author did a comprehensive work in searching and screening the eligible studies, evaluate the quality through different aspects, and the revision has clarified my previous questions.